

IMAGE BY **HARDIK CHHABRA/AI**

country towards fuller semiconductor capabilities.

The same pattern is visible in spacetech. The report counts more than 140 start-ups working across India's spacetech value chain. It attributes the growth to the opening-up of the sector, ISRO's encouragement to private participation and the opportunity created by a growing commercial space economy.

ENGINEERING TALENT

For AGNIT co-founder and CEO Hareesh Chandrasekar, this is not merely another start-up cycle. "We

are in the middle of a shift from mainly a service-driven economy to an economy that actually owns IP (intellectual property)," he says.

India's strength in engineering has never really been in question. The problem was where that talent sat in the value chain.

What is changing is the willingness—and increasingly the ability—to turn that expertise into Indian-owned products.

"Pre-2021-2022, nobody was even interested in semiconductors," says Chandrasekar. "That changed once the ISM and DLI schemes were announced to

THE TECH SHIFT

➤ A new generation of start-ups is building indigenous IP in semiconductors, propulsion, space, aerospace and energy, transforming India from being an assembler or technology integrator to a technology owner

➤ Semiconductor incentives, space-sector reforms and support for deep-tech R&D have given entrepreneurs greater confidence to enter capital-intensive, strategic industries and build products that were previously dominated by government institutions or global companies

➤ Building a prototype is only the first step. Start-ups now need specialised suppliers, certifications, manufacturing capacity, patient capital and customers to turn indigenous technology into commercially viable products at scale

➤ India has around 20% of global chip-design talent and more than 140 spacetech start-ups, but gaps in advanced components, certification and industrial automation remain stark—the country has only about four robots per 10,000 manufacturing workers versus roughly 400 in China

➤ Government and defence can provide critical early customers, but deep-tech start-ups must eventually win private and international markets. With India still estimated to be about 15 years behind in deep tech, the real journey is from Invent in India to Compete From India